

The Chair of Operations Management, in cooperation with Munich Airport, is offering a

Project Study / Interdisciplinary Project

on the topic

Developing a Model for Dynamically Determining Routes and Travel Times on the Airfield of Munich Airport

Scope

Knowledge of the distances and travel times between aircraft handling positions is necessary for the efficient execution of ground handling processes at an airport. A previous study developed an app that calculates the distance and static travel time for vehicles travelling between any two locations on Munich Airport's airfield. However, travel times are not static. On some routes, vehicles must pass through sections of the apron that are used by aircraft, which have right of way. During peak periods with high aircraft traffic, there are more aircraft crossings at these sections, leading to higher expected waiting times and thus increased overall travel times.

This follow-up study will determine time-dependent travel times that consider waiting times due to aircraft crossings. To achieve this, two approaches will be undertaken, resulting in two project studies.

Project 1: Aircraft crossings at each section are determined based on the flight schedule and apron layout. Crossing frequencies can be determined either deterministically or by simulation using the CAST Aircraft simulation tool.

Project 2: Aircraft crossings are determined based on a database containing records of past aircraft movements.

Based on the frequencies derived by each project, the existing app will be extended to include time-dependent waiting times, and thus travel times, for each origin-destination pair.

Requirements

- Interest in air traffic and airports
- Knowledge of the fundamentals of Operations Research
- Basic knowledge in Simulation (e.g., through the module MOS) and/or database systems
- Good IT and programming skills, preferably Python
- Reliability and resilience

Start and duration: Immediately; duration: 3 months full-time, longer for part-time work

Language: German or English

Number of participants for each project study: 2-4

Supervisor: Andreas Hagn (MSc.)

Supervisor Airport Munich: Martin Bertling

Contact: If you are interested, please contact andreas.hagn@tum.de